Ankur Singh

LinkedIn: linkedin.com/in/ankursingh77/ **Email:** ankurrajput2118@gmail.com
GitHub: github.com/ankursingh23 **Phone:** +91-8604348068

Skills

- **Languages**: Python, C++, SQL
- **Frameworks/Tools**: Django, Tkinter, MySQL, VSCode, Jupyter, Pandas, NumPy, Git
- **Core CS Concepts**: Data Structure & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management System (DBMS), Operating System (OS)
- **Soft Skills**: Problem Solving, Persistence, Analytical Thinking, Adaptability

Projects

Dynamic Heterogeneous Data Structure | C++, OOP, STL | GitHub Jul-Aug 2025

- Designed and constructed a custom heterogeneous data structure in C++ capable of storing multiple data types with type safety, high-performance operations, and intuitive Python-like usability.
- Developed a custom heterogeneous data structure using enum-based type tracking, OOPs principles, STL components, and a doubly linked list, enabling optimized insertion, deletion, traversal, and type-safe data handling.
- Formulated an efficient indexing system with unordered_map and built an interface with operator overloading and dynamic memory management, achieving flexible data manipulation with high performance.
- Achieved a high-performance, type-safe data structure with O(N) average operational complexity, enabling flexible heterogeneous data handling and intuitive user-level interactions

File Compression and Decompression (Huffman Coding) | GitHub Feb-Mar 2025

- Implemented and engineered a Huffman Coding system capable of performing efficient, lossless file compression and decompression with high speed, modularity, and scalability.
- Built a complete Huffman coding pipeline using priority queue (heapq), bit manipulation, and optimized file I/O for accurate and fast encoding/decoding.
- Architected a modular, reusable, and scalable architecture with cleanly separated encoding/decoding modules, enabling binary data handling and future system integration.
- Delivered up to 60% reduction in file size (depending on data type) with a maintainable, high-performance compression system supporting reliable, lossless data reconstruction.

Certificates

- GFG 160 – 160 Days of Problem Solving (DSA & Algorithms) | GeeksForGeeks | Link — Nov 2025
- SQL (Intermediate) |HackerRank| Link — Sep 2025
- Certified in Django Framework (Self-Paced) |GeeksForGeeks | Link — Aug 2025

Achievements

- Ranked 32nd on GeeksforGeeks LPU coding leaderboard—recognized for strong algorithmic problem solving and data structure proficiency.
- Solved 1000+ algorithmic and DSA problems across GeeksforGeeks, LeetCode, and Code Studio, demonstrating consistent coding proficiency and analytical thinking.

Education

Lovely Professional University Punjab, India
Bachelor of Technology (Hons.) – Computer Science and Engineering | CGPA: 8.9 Aug 2023- Present

Police Modern School Lucknow, UP
Intermediate | Percentage: 84 Apr 2021 - Jun 2022

Police Modern School Lucknow, UP
Matriculation | Percentage: 88 Apr 2019 – Mar 2020